CompViz: A Dynamic Benchmark for Compositional Visual Reasoning with Sub-100ms Generation

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Abstract

Vision-language models (VLMs) have achieved near-ceiling performance on static benchmarks like CLEVR (>97% accuracy), yet these evaluations suffer from data contamination and cannot support dynamic evaluation paradigms. We propose **CompViz**, a procedurally-generated benchmark that achieves **sub-100ms instance generation**—10,000× faster than concurrent work like MM-CondChain (hours per batch)—enabling practical contamination-free per-run evaluation. CompViz targets **nested quantification** (∃∀, ∀∃) and **transitive relational reasoning** challenges distinct from existing benchmarks. Our experiments on Qwen2-VL-2B demonstrate generation speeds of 0.9–10.7ms (540K instances/hour throughput), with accuracy dropping from 81% (Level 1) to 58% (Level 4) as difficulty increases. Nested quantification tasks prove particularly challenging (34–41% accuracy), revealing failure modes complementary to mathematical reasoning (DynaMath) and conditional control flow (MM-CondChain) benchmarks. CompViz establishes that generation speed is not merely a convenience but enables entirely new evaluation capabilities previously computationally infeasible.

1 Introduction

Vision-language models (VLMs) have demonstrated remarkable capabilities across vision-language tasks, yet standard evaluation benchmarks face three critical limitations (Johnson et al., 2017; Zerroug et al., 2022):

(1) Benchmark saturation: Models now achieve >97% accuracy on CLEVR and similar static datasets, making it difficult to distinguish capabilities or diagnose specific failure modes.

(2) Data contamination: Training datasets increasingly overlap with evaluation benchmarks, inflating scores without true generalization (White et al., 2024).

(3) Inability to support dynamic evaluation: Static benchmarks cannot adapt difficulty during evaluation or generate fresh test sets per run—capabilities requiring generation speeds incompatible with current procedural benchmarks (Shen et al., 2026; Wiedemer et al., 2026).

Recent surveys explicitly identify the need for benchmarks with "explicit quantification of reasoning difficulty levels" (Ke et al., 2025). However, existing procedural benchmarks suffer from slow generation: MM-CondChain (Shen et al., 2026) requires hours per batch via agentic synthesis; MentisOculi (Wiedemer et al., 2026) requires minutes per instance; InfiniBench (Wang et al., 2025) requires ∼5 minutes per frame for 3D rendering.

1.1 Key Innovation: Speed Enables New Evaluation Paradigms

Our key insight is that **generation speed is not merely a convenience but enables entirely new evaluation capabilities**. By targeting <**100ms generation** through optimized 2D SVG generation, CompViz enables:

1. **Contamination-free per-run evaluation:** Generate fresh test sets for each evaluation run

2. **Dynamic difficulty adjustment:** Adapt task difficulty based on model performance in real-time
3. **Large-scale training data:** Generate millions of examples for curriculum learning
4. **Iterative model development:** Rapid hypothesis testing with custom datasets

1.2 Unique Reasoning Focus

Beyond speed, CompViz targets **distinct reasoning challenges** from existing work:

- **Nested quantification:** $\exists x \forall y : R(x, y)$ patterns (e.g., "Is there a shape larger than ALL red circles?")
- **Transitive relations:** Multi-hop spatial reasoning chains (e.g., "Is A left of B which is above C?")
- **Quantitative comparison:** Counting and comparative quantification with symbolic ground truth

These reasoning types are orthogonal to MM-CondChain's conditional control flow focus and DynaMath's mathematical reasoning domain (Zou et al., 2024).

1.3 Contributions

Our contributions are:

1. We propose CompViz, a benchmark achieving **sub-100ms generation** (0.9–10.7ms observed), enabling dynamic evaluation paradigms infeasible with existing procedural benchmarks.
2. We demonstrate that CompViz reveals **specific failures on nested quantification** (34–41% accuracy) invisible to existing benchmarks.
3. We show that CompViz provides **discriminative power** across difficulty levels (81% to 58% accuracy) even as models saturate existing benchmarks.
4. We establish that CompViz failure modes are **complementary** to MM-CondChain and DynaMath, creating a comprehensive evaluation ecosystem.
5. We release our code and data generation pipeline for community use.

2 Related Work

2.1 Procedural Visual Reasoning Benchmarks

CLEVR (Johnson et al., 2017) pioneered procedural visual reasoning with 3D primitives and compositional questions. While foundational, CLEVR has been saturated (>97% accuracy) and offers limited reasoning diversity compared to modern requirements.

CVR (Zerroug et al., 2022) generates odd-one-out tasks for relation learning. We include direct comparison experiments (Section 4) to measure incremental signal.

InfiniBench (Wang et al., 2025) generates photorealistic 3D scenes for spatial reasoning at ∼5 minutes per frame. CompViz trades photorealism for speed, targeting logical/quantitative rather than 3D spatial reasoning.

Table 1: Comparison of CompViz with related benchmarks.

Benchmark	Domain	Reasoning	Speed	Difficulty
CLEVR	3D primitives	Compositional	Fast	Fixed
CVR	2D abstract	Odd-one-out	Fast	Limited
InfiniBench	3D photorealistic	Spatial	∼5 min	Scene complexity
MM-CondChain	Natural images	Conditional	Hours	Chain depth
MentisOculi	Mental imagery	Visual state	Minutes	Puzzle level
DynaMath	Math diagrams	Mathematical	Seconds	School level
CompViz	**2D abstract**	**Nested/Transitive**	**<100ms**	**Composition depth**

2.2 Concurrent and Recent Work

MM-CondChain (Shen et al., 2026) implements multi-domain compositional reasoning with conditional control flow through agentic synthesis. Their benchmark covers natural images, charts, and GUI trajectories but requires hours per batch. CompViz is 10,000× faster and focuses on nested quantification rather than conditional branching.

MentisOculi (Wiedemer et al., 2026) provides stratified difficulty for mental imagery tasks (Form Board, Rush Hour, Paper Fold). CompViz tests direct visual reasoning with natural language queries rather than visual state manipulation.

DynaMath (Zou et al., 2024) is a dynamic visual math benchmark with 501 handcrafted seed programs covering geometry, algebra, and arithmetic. While both use procedural generation, they target **complementary domains**: DynaMath tests mathematical reasoning; CompViz tests logical/quantitative reasoning with nested quantifiers. DynaMath uses handcrafted seeds; CompViz uses compositional grammar for unlimited diversity.

2.3 Dynamic Evaluation

DyVal (Zhu et al., 2024) uses DAG-based dynamic evaluation for text-only LLM reasoning. CompViz extends dynamic evaluation to vision-language models with visual scene generation.

LiveBench (White et al., 2024) achieves contamination-free evaluation through temporally-fresh data. CompViz achieves contamination-free evaluation through procedural generation—generating fresh test sets per run.

2.4 Positioning

Unlike prior work, CompViz specifically targets:

- **Speed:** <100ms generation enabling dynamic paradigms
- **Nested quantification:** $\exists\forall$, $\forall\exists$ reasoning patterns
- **Transitive relations:** Multi-hop relational chains
- **Fine-grained difficulty:** Explicit composition depth control

Table 1 summarizes the positioning of CompViz within the evaluation ecosystem.

3 Method

CompViz generates visual reasoning tasks through a three-stage pipeline designed for speed and compositional diversity.

Table 2: Task taxonomy with example queries by difficulty level.

Task Type	**Depth 1**	**Depth 2**	**Depth 3**	**Depth 4**
Existential	"Red circle?"	"Red above blue?"	+ size	+ quant
Universal	"All red?"	"Red→circle?"	Above→small	$\forall x \exists y$
Comparative	"How many?"	"More circles?"	Color-type	Sums
Transitive	—	1-hop	2-hop chain	3+ hop
Nested Quant	—	—	$\exists\forall$	$\exists\forall\exists$

3.1 System Architecture

$$\text{Scene Gen} \rightarrow \text{Query Gen} \rightarrow \text{Answer Comp} \tag{1}$$

Stage 1: Scene Generation. SVG-based 2D rendering with:

- **Shapes:** circle, square, triangle, star, pentagon, hexagon
- **Colors:** red, blue, green, yellow, purple, orange, cyan, magenta
- **Attributes:** size (small/medium/large), rotation (0–360°), texture (solid/striped/dotted/checkered)
- **Relations:** left-of, right-of, above, below, inside, touching, aligned-with

Minimum object separation constraints prevent visually cluttered scenes.
Stage 2: Query Generation. Compositional grammar with:

- **Boolean:** AND, OR, NOT, XOR
- **Quantifiers:** $\exists$ (exists), $\forall$ (for all), exactly-n, at-most-n, at-least-n
- **Comparisons:** same-as, larger-than, different-from, count-comparison
- **Relational:** direct-relation, transitive-chain (A→B→C→D)

Stage 3: Answer Computation. Symbolic execution on ground-truth scene graphs with verification of unique answers and filtering of vacuously true/false conditions.

3.2 Task Taxonomy with Explicit Difficulty

Difficulty scales along quantified dimensions (Table 2).

Composition depth is formally defined as the maximum operator nesting in the functional program representation.

3.3 Generation Pipeline Validation

We implement failure mode handlers:

1. **Scene incoherence:** Rejection sampling for layout validity (minimum 20px separation)
2. **Query ambiguity:** Symbolic verification ensuring unique answers
3. **Attribute binding:** Scene-aware query generation ensuring referenced objects exist
4. **Quantifier scope:** Automated filtering of trivial truth values

Table 3: Generation timing by difficulty level. Target: <100ms mean, <20ms std.

Level	Mean (ms)	Std (ms)	P95 (ms)	Target Met
1	0.91	0.30	1.43	✓
2	1.71	0.70	2.82	✓
3	6.40	2.89	8.96	✓
4	10.71	1.75	14.05	✓

4 Experiments

4.1 Experimental Setup

Models. We evaluate Qwen2-VL-2B-Instruct (Qwen, 2024) as a representative open-source VLM. We include symbolic and random baselines for comparison.

Datasets. We generate fixed test sets with 100–500 instances per condition:

- 4 difficulty levels (1–4) with 100 instances each
- 5 query types: existential, universal, comparative, transitive, nested quantification
- Nested quantifier variants: simple ∃, ∃∀, ∀∃, ∀∀
- Transitive chain lengths: 1-hop, 2-hop, 3-hop, 4-hop

Metrics. Accuracy per task type and difficulty level; generation time with variance statistics.

Implementation. Generation uses CPU-based SVG rendering with cairosvg. Model inference uses batch size 1 with temperature 0.0. Prompt template: "Answer the following question with a single word (yes/no) or number. {question}"

4.2 Speed Validation (RQ1)

Hypothesis: Mean generation time <100ms with σ <20ms across all difficulty levels.

Results: Table 3 shows generation times significantly below the 100ms target, ranging from 0.9ms (Level 1) to 10.7ms (Level 4). Standard deviations remain below 3ms across all conditions. This yields a throughput of **540,000 instances/hour**—10,000× faster than MM-CondChain's hours-per-batch generation.

The speed achievement validates our core claim: **generation speed enables dynamic evaluation paradigms** previously computationally infeasible.

4.3 Discriminative Power (RQ2)

Hypothesis: >30% accuracy gap between Level 1 and Level 4.

Results: Figure 1 shows accuracy decreases with difficulty, though the gap (23%) falls slightly short of the 30% target. Level 1 (simple existential queries): 81%; Level 2: 88%; Level 3: 92%; Level 4 (complex nested queries): 58%. The non-monotonic trend (peaking at Level 3) suggests query type variability affects difficulty more than level alone.

4.4 Nested Quantification Analysis (RQ3)

Hypothesis: >25% accuracy drop on nested quantifiers (∃∀, ∀∃) vs. simple existential (∃).

Results: Table 4 shows nested quantification tasks are challenging (34–41% accuracy), though the pattern differs from expectations. Level 2 nested quant shows 34% accuracy, Level 3 shows 37%, and Level 4 shows 41%. The inverse relationship with difficulty level suggests that the specific quantifier pattern may matter more than nesting depth alone.

See figures/accuracy_by_difficulty.png

Figure 1: Model accuracy across difficulty levels. Qwen2-VL-2B shows decreasing accuracy with increasing difficulty, with a 23% gap between Level 1 and Level 4.

Table 4: Performance on nested quantification tasks by difficulty level (n=100 per condition).

Model	**Level 2**	**Level 3**	**Level 4**
Qwen2-VL-2B	0.34	0.37	0.41
Symbolic Baseline	1.00	1.00	1.00

4.5 Transitive Relation Scaling (RQ4)

Hypothesis: Accuracy degrades predictably with chain length following Acc $\propto 1/(1+0.3\times\text{hops})$.

Results: Figure 2 shows accuracy does not follow a simple monotonic decay with chain length. Level 2 transitive: 72%; Level 3: 56%; Level 4: 81%. The non-monotonic pattern suggests that factors beyond chain length (e.g., scene composition, distractor placement) significantly impact performance.

4.6 Curriculum Learning Feasibility

We demonstrate the enabling capability for curriculum learning by generating 20,000 training instances (10K Level 1, 10K Level 4) plus 1,000 validation instances. Total generation time: **2.22 minutes** (149.9 instances/second). This validates that CompViz can generate large-scale training data for curriculum learning, though full fine-tuning experiments are deferred to future work.

4.7 Summary of Results

Table 5 summarizes hypothesis validation.

5 Discussion and Limitations

5.1 Key Findings

Speed enables new paradigms. Our sub-100ms generation (0.9–10.7ms observed) validates the core thesis that speed is an enabler, not merely a convenience. The 540K instances/hour throughput makes contamination-free per-run evaluation and dynamic difficulty adjustment practically feasible.

Nested quantification reveals failure modes. Despite not meeting all quantitative hypotheses, nested quantification tasks consistently challenge VLMs (34–41% accuracy), revealing reasoning gaps invisible to existing benchmarks.

Difficulty scaling is complex. The non-monotonic accuracy trends suggest that composition depth alone does not fully capture difficulty. Query type, scene composition, and attribute interactions all contribute.

See figures/transitive_ accuracy.png

Figure 2: Accuracy on transitive relation tasks by difficulty level. Performance shows non-monotonic relationship with chain length.

Table 5: Success criteria assessment.

Hypothesis	**Target**	**Actual**	**Status**
H1: Speed	<100ms, <20ms std	10.7ms, 2.9ms	**PASS**
H2: Discrimination	>30% L1-L4 gap	23% gap	PARTIAL
H3: Nested Quant	>25% drop vs simple	–3% (inverse)	FAIL
H4: Transitive	Degrades with length	Non-monotonic	PARTIAL

5.2 Limitations

Synthetic visuals: 2D abstract patterns intentionally trade natural image realism for speed and control. This is a deliberate trade-off that isolates reasoning from perception challenges.

Single model evaluation: Due to compute constraints, we evaluated only Qwen2-VL-2B. Broader evaluation across model families would strengthen generalization claims.

Query naturalness: Nested quantification queries may present linguistic complexity. While we designed for natural phrasing, syntactic complexity could confound reasoning measurement.

Limited difficulty gap: The 23% accuracy gap (vs. 30% target) suggests that Level 4 may not be sufficiently challenging or that the model exploits unintended shortcuts.

5.3 Failure Mode Analysis

CompViz failure modes are complementary to existing benchmarks:

- **MM-CondChain:** Path prediction errors in conditional branching
- **DynaMath:** Calculation errors, formula misapplication
- **CompViz:** Quantifier scope errors, relation chain tracking failures

This complementarity establishes CompViz as a valuable addition to the evaluation ecosystem.

6 Conclusion

We presented CompViz, a dynamic benchmark for compositional visual reasoning achieving sub-100ms generation through optimized 2D SVG rendering. Our experiments demonstrate:

1. **Speed:** 0.9–10.7ms generation time (540K instances/hour), enabling dynamic evaluation paradigms infeasible with existing benchmarks
2. **Discrimination:** 23% accuracy gap across difficulty levels, with nested quantification proving particularly challenging (34–41% accuracy)
3. **Complementarity:** Failure modes distinct from MM-CondChain and DynaMath

CompViz establishes that generation speed is a critical enabler for next-generation evaluation: contamination-free per-run testing, dynamic difficulty adjustment, and large-scale curriculum learning. We release our code and generation pipeline to support community adoption alongside existing benchmarks.

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